Simple SST forecast baselines in two North Pacific boxes: a test-period evaluation

Question. How hard are simple baselines to beat for 1-, 7- and 30-day SST forecasts, and does a per-grid-point Ridge regression help? This is a baselines study, not an ocean forecasting model.

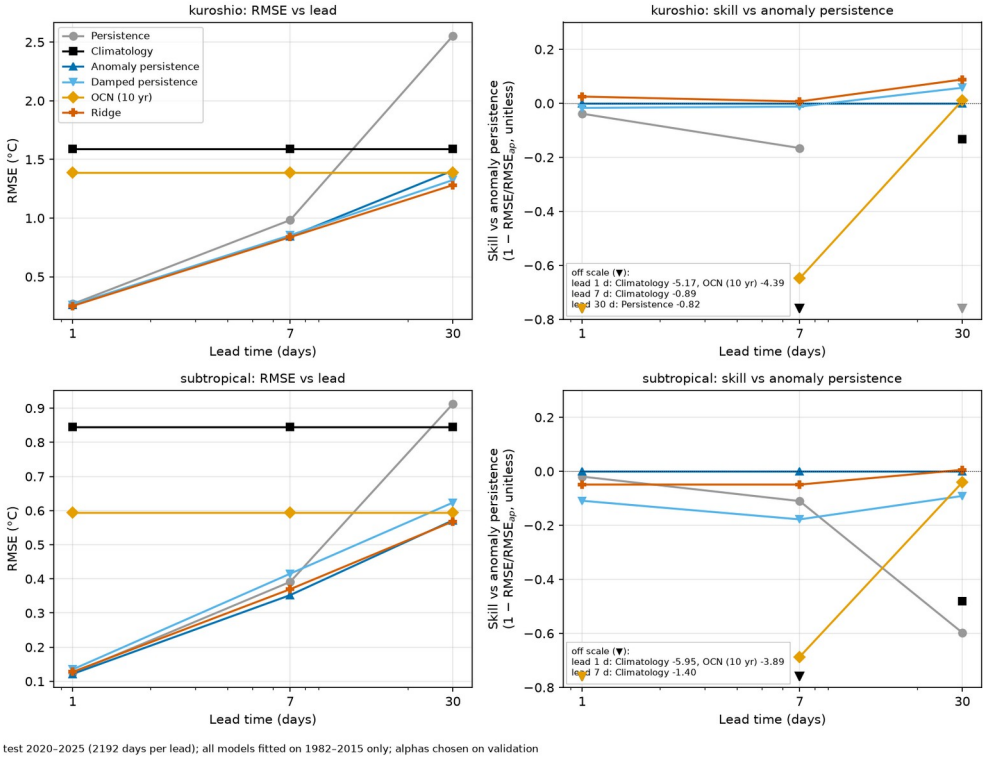
Data. NOAA OISST v2.1 daily 0.25° SST, two 40x40-point boxes (Kuroshio extension, per-point anomaly std 1.13 °C; subtropical, 0.69 °C), 1982-01-01–2025-12-31 (16,071 days). ERDDAP is the primary source; 1,194 days it lacks were filled from NOAA PSL after checks (double fetch, range, step, exact match on overlaps: max |difference| 3.8e-06 °C).

Protocol. Training 1982–2015 (climatology: 3 harmonics; Ridge fit), validation 2016–2019 (Ridge alpha), test 2020–2025 (2192 days per lead) evaluated once after a frozen, tagged configuration. Score: cos(lat)-weighted RMSE; skill = 1 − RMSE/RMSE of anomaly persistence.

Baselines. persistence; climatology; anomaly persistence; damped persistence (training-year lag correlation); OCN (mean of the same day over the previous 10 years); Ridge (anomaly lags 0–6 and 30-day mean anomaly, one model per grid point and lead).

Test results: RMSE in °C (skill vs anomaly persistence).

Method	kuros. 1 d	kuros. 7 d	kuros. 30 d	subtr. 1 d	subtr. 7 d	subtr. 30 d
Persistence	0.268 (-0.039)	0.983 (-0.166)	2.554 (-0.817)	0.124 (-0.019)	0.391 (-0.110)	0.913 (-0.597)
Climatology	1.591 (-5.174)	1.591 (-0.886)	1.591 (-0.132)	0.846 (-5.955)	0.846 (-1.399)	0.846 (-0.480)
Anomaly persistence	0.258 (+0.000)	0.843 (+0.000)	1.405 (+0.000)	0.122 (+0.000)	0.353 (+0.000)	0.571 (+0.000)
Damped persistence	0.262 (-0.017)	0.854 (-0.012)	1.324 (+0.058)	0.135 (-0.109)	0.415 (-0.178)	0.623 (-0.091)
OCN (10 yr)	1.390 (-4.394)	1.390 (-0.648)	1.390 (+0.011)	0.595 (-3.890)	0.595 (-0.687)	0.595 (-0.041)
Ridge	0.251 (+0.025)	0.838 (+0.007)	1.281 (+0.088)	0.128 (-0.049)	0.370 (-0.049)	0.568 (+0.006)



test 2020–2025 (2192 days per lead); all models fitted on 1982–2015 only; alphas chosen on validation

Key findings.

1. **Anomaly persistence is the bar to beat, especially at short range.** At 1 day all anomaly-based methods have RMSE of 0.122–0.258 °C (subtropical–Kuroshio, anomaly persistence), while climatology gives 0.846–1.591 °C and OCN 0.595–1.390 °C. Ridge's margin over anomaly persistence at 1 day is small (skill +0.025 Kuroshio, -0.049 subtropical).
2. **Ridge helps in the high-variability Kuroshio box, not reliably in the subtropical box.** Kuroshio: Ridge beats anomaly persistence at every lead (skill +0.025, +0.007, +0.088 at 1, 7, 30 d). Subtropical: Ridge is worse at 1 and 7 days (-0.049, -0.049) and about equal at 30 d (+0.006); on validation it had been better at all leads (+0.022, +0.056, +0.090). The validation advantage did not carry over.
3. **Plain persistence fails at 30 days, and the warm shift hurts climatology-based forecasts.** Persistence at 30 d has skill -0.817 (Kuroshio) and -0.597 (subtropical) relative to anomaly persistence. Against the 1982–2015 climatology the test years are warm: mean bias of the climatology forecast is -0.827 °C (Kuroshio) and -0.656 °C (subtropical); Ridge at 30 d is also biased low (-0.442, -0.287 °C).

Limitations. Two small regions; simple statistical baselines; OISST is a smooth interpolated analysis; hybrid ERDDAP + PSL data (verified on overlaps); v2/v2.1 label change at 2016; warm shift vs the 1982–2015 climatology, not detrended; Ridge ignores spatial structure; its alpha sits at the grid's lower edge in several cases; one fixed split.

Next step. Optionally a small LSTM under the same protocol (reference: Hao et al. 2025, IEEE Trans. Cybernetics 55(6), review). Code and data checks: README.md, results/*.json.